Journal of Drug Targeting, 2010; 18(4): 321–331

informa
healthcare

RESEARCH ARTICLE

Mucoadhesive chitosan microspheres of carvedilol for nasal administration

Sanjay Patil[1], Anil Babbar[2], Rashi Mathur[2], Anil Mishra[2], and Krutika Sawant[1]

[1]*New Drug Delivery Systems Laboratory, TIFAC Center of Relevance and Excellence in NDDS, Pharmacy Department, The M. S. University of Baroda, Fatehgunj, Vadodara-390002, Gujarat, India, and* [2]*Institute of Nuclear Medicine and Allied Sciences, Ministry of Defense, Government of India, Delhi 110054, India*

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

Abstract

The aim of the present study was to develop and characterize chitosan mucoadhesive microspheres of carvedilol (CRV) for nasal delivery to improve bioavailability for treatment of hypertension and angina pectoris. The microspheres were prepared by emulsification-cross-linking method and evaluated for size, shape, entrapment efficiency (EE), *in vitro* mucoadhesion, *in vitro* drug release, differential scanning calorimetry (DSC) and X-ray diffraction (XRD). The mucoadhesive properties were also evaluated by Freundlich and Langmuir adsorption isotherms. *In vivo* tests were carried out in rabbits. The microspheres were spherical with size of 20–50 µm, which is favorable for intranasal absorption. The EE was observed from 42% to 68% while percentage mucoadhesion was from 74% to 88%. A strong interaction between mucin and chitosan microspheres was detected explaining adsorption with electrostatic interaction. The microspheres released around 75% of drug in 8 h. DSC and XRD studies revealed that CRV was molecularly dispersed. The absorption rate was rapid and the absolute bioavailability was high, 72.29%. The gamma scintigraphy indicated that the microspheres cleared slowly from the nasal cavity. It was concluded that chitosan microspheres could be used to deliver CRV following nasal administration for improving the bioavailability.

Keywords: *Microspheres; chitosan; CRV; nasal delivery; pharmacokinetics*

Copyright © 2010 Informa UK Limited
Not for Sale or Commercial Distribution
Unauthorized use prohibited. Authorized users can download, display, view and print a single copy for personal use

Introduction

The nasal route has gained tremendous attention for systemic drug delivery by many researchers within the last few decades due to its great potential utility for drug delivery. It offers an attractive alternative for drugs that have limited oral bioavailability, are destroyed by gastrointestinal fluids, or are highly susceptible to hepatic first pass or gut wall metabolism. Nasal drug delivery also offers the convenience and safety of being non-invasive (Chien et al., 1989). In addition, nasal drug administration results in quick onset of action as compared to oral, sublingual and transdermal administrations. The nasal cavity as a site for the systemic absorption of drugs has some advantages which include relatively large surface area, porous endothelial basement membrane, highly vascularized epithelial layer, high total blood flow per cm^3, avoiding the first pass metabolism and easy access (Cornaz and Buri, 1994; Mygind and Dahl, 1998). Nasal mucociliary clearance is one of the most important limiting factors for nasal drug delivery. It severely limits the time allowed for drug absorption to occur and effectively rules out sustained nasal drug administration. However, mucoadhesive preparations have been developed to increase the contact time between the dosage form and mucosal layers of nasal cavities, thereby enhancing drug absorption (Vidgren et al., 1991; Ugwoke et al., 2000). Thus, mucoadhesive microspheres have been developed to decrease the effect of mucociliary clearance (Illum et al., 1987). The microspheres form a gel like layer,

Address for Correspondence: Sawant Krutika K, Professor of Pharmaceutics, Pharmacy Department, Faculty of Technology and Engineering, The Maharaja Sayajirao University of Baroda, Kalabhavan, Vadodara-390 001, Gujarat, India. E-mail: dr_krutikasawant@rediffmail.com

(Received 06 October 2009; revised 27 December 2009; accepted 29 January 2010)

ISSN 1061-186X print/ISSN 1029-2330 online © 2010 Informa UK Ltd
DOI: 10.3109/10611861003663523

RIGHTSLINK

which is cleared slowly from the nasal cavity, resulting in a prolonged residence time of the drug formulation. It has been established that mucoadhesive microspheres are able to significantly increase the systemic absorption of conventional drugs as well as polypeptides across the nasal membrane without the use of absorption enhancing agents that have the potential for irritation or damage (Illum, 1998).

Drugs cross the nasal mucosal membrane either transcellularly (across the cells) or paracellularly (between the cells). Lipophilic drugs are transported transcellularly by a concentration dependent passive diffusion process, by facilitated diffusion using a receptor or carrier molecule, or by vesicular transport mechanisms (Illum, 2002).

Carvedilol (CRV) is a non-selective β-adrenergic antagonist used in the treatment of hypertension and angina pectoris (Packer et al., 2002). It is well absorbed from the gastrointestinal tract but is subject to considerable first-pass metabolism in the liver; its oral absolute bioavailability is about 25% (Thummel and Shen, 2001).

Chitosan, a linear polysaccharide derived by the deacetylation of chitin obtained from crustacean shells, has emerged as a useful drug delivery matrix because of it polycationic nature, biodegradability, biocompatibility, mucoadhesiveness and ease of physical and chemical modification. The interaction between cationic amino groups on chitosan and anionic moieties such as sialic and sulfonic acids on the mucus layer is responsible for its mucoadhesiveness (Bernkop-Schnurch et al., 2003). In addition, chitosan enhances epithelial permeability through the opening of tight junctions between epithelial cells (Fernandez-Urrusuno et al., 1999). Chitosan based particulate delivery system can reduce the clearance rate from the nasal cavity, thereby increasing the contact time of the delivery system with the nasal mucosa (Soane et al., 1999). Chitosan microparticles can be prepared using a covalent cross linker like glutaraldehyde (GA) and such microparticles are widely used as a delivery system (Thanoo et al., 1992; Blanco et al., 2000; Wang et al., 1996). Other methods of microsphere preparation like spray drying (Mi et al., 1999; Gavini et al., 2008) and ionic gelation (Berthod et al., 1994; van der Lubben et al., 2003) using sodium sulphate or tripolyphosphate are reported.

The objective of the present study was preparation of mucoadhesive chitosan microspheres for nasal administration of CRV. The microspheres were prepared by emulsification-cross-linking method. The microspheres were characterized in terms of particle size, morphology, entrapment efficiency (EE), *in vitro* mucoadhesion, *in vitro* drug release, X-ray diffraction (XRD) and differential scanning calorimetry (DSC). *In vivo* tests were performed on rabbit which is one of the preferred animal models for pharmacokinetic studies in nasal drug delivery.

Experimental

Materials

Chitosan was a gift sample from Ample Effect Sdn Bhd, Selangor (Malaysia) and used without any modification and purification (molecular weight < 600,000 Da, degree of deacetylation > 85%). CRV was a gift sample from Torrent Research Centre, Ahmedabad, India. Liquid paraffin (light and heavy), glutaraldehyde (25% aqueous solution), and hexane were purchased from S. D. Fine Chemicals, Mumbai, India. Dioctyl sodium sulfosuccinate (DOSS) was procured from Wilson Laboratories, Mumbai, India. Mucin (type III, partially purified from porcine stomach, bound sialic acids ~ 1%) and stannous chloride dihydrate ($SnCl_2.2H_2O$) were purchased from Sigma Chemical Company, St. Louis, MO. Sodium pertechnetate, separated from molybdenum-99 (99m) was provided by Regional Center for Radiopharmaceutical Division (northern region), Board of Radiation and Isotope Technology, New Delhi, India. All other chemicals and reagents used in the study were of analytical grade.

Methods

Preparation of chitosan microspheres

Chitosan microspheres were prepared by simple w/o emulsification-cross-linking process using liquid paraffin (heavy and light, 1:1) as external phase (Thanoo et al., 1992). Briefly, chitosan (0.2 g) was dissolved in 2% aqueous acetic acid solution (10 mL) by continuously stirring until a homogeneous solution was obtained. The drug (0.1 g) was added in chitosan solution and the dispersion was added slowly through syringe to 100 mL of liquid paraffin (heavy and light, 1:1) containing 0.2% w/v of DOSS as stabilizer under constant stirring at 1200 rpm for 15 min using a Eurostar (IKA Labortechnik, Germany) high speed stirrer. To this W/O emulsion, 1 mL of GA (25% solution, as cross-linking agent) was added slowly and stirring was continued for 2 h. The hardened microspheres were separated by vacuum filtration and washed several times with hexane to remove oil. Finally, microspheres were washed with distilled water to remove unreacted GA. The microspheres were air dried for 24 h and then stored in vacuum desiccator until further use. The formula for the various batches of microspheres is shown in Table 1.

Particle size measurements

Particle size and size distribution of the microspheres were determined by laser light scattering on a Malvern particle size analyzer (Malvern Mastersizer 2000, Malvern Instruments, UK). The dispersion of microspheres was added to the sample dispersion unit containing the stirrer

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

Table 1. Various formulation parameters used in the preparation of microspheres.

Formulation code	Drug: polymer ratio	Volume of GA (mL)	Cross-linking time (h)
CHC1	1:2	1	1
CHC2	1:1	1	1
CHC3	1:2	2	1
CHC4	1:1	2	1
CHC5	1:2	1	2
CHC6	1:1	1	2
CHC7	1:2	2	2
CHC8	1:1	2	2

and stirred in order to reduce the aggregation between the microspheres and the laser obscuration range was maintained between 5% and 15%. The average volume mean particle size was measured after performing the experiment in triplicate.

Morphological characterization

All batches of microspheres were preliminarily checked for shape and size by optical microscopy. A light transmission microscope (Olympus Optical Co. Ltd., Japan) at 10X magnification was employed. The samples were prepared by suspending a small amount of microspheres in paraffin oil. The surface morphology of the CRV loaded microspheres was examined by scanning electron microscopy (SEM; JSM 5610 LV, Jeol Datum Ltd., Japan). The samples were mounted directly onto the SEM sample holder using double-sided sticking tape and images were recorded at the required magnification at the acceleration voltage of 5 kV.

Flow properties

Flow properties of the microspheres were evaluated by determining the angle of repose and compressibility index (Banker et al., 1991; Taylor et al., 2000). The fixed funnel method was used to find out angle of repose. Compressibility index values of the microspheres were determined by measuring the initial volume V_0 and final volume V after subjecting to 100 tappings in a graduated measuring cylinder using the following equation:

$$\%\ \text{compressibility index} = (1 - \frac{V}{V_0}) \times 100$$

Determination of EE

The CRV-loaded microspheres (5 mg) were added to a mixture of 10 mL of methanol and 0.1N HCl (3:2) under stirring. The mixture was filtered and the amount of CRV was determined spectrophotometrically at 242 nm on UV spectrophotometer (Shimadzu UV1610, Japan). It was confirmed from preliminary UV studies that the presence of dissolved polymers did not interfere with the absorbance of the drug at 242 nm. The drug loading and percentage EE was calculated using following equations:

$$\text{Drug loading} = \frac{\text{Actual drug content of weighed quantity of microspheres}}{\text{Weighed quantity of microspheres}}$$

$$\%\ \text{Entrapment efficiency} = \frac{\text{Actual loading}}{\text{Theoretical loading}} \times 100$$

Adsorption studies

Mucus glycoprotein assay. A periodic acid/Schiff colorimetric method reported by Mantle and Allen (Mantle and Allen, 1978) was used to determine the free mucin concentration in order to assess the amount of mucin adsorbed on the chitosan microspheres and its effect on the assessment of mucoadhesive behavior of chitosan microspheres. Two reagents were prepared. Schiff reagent contained 100 mL of 1% basic fuchsin (pararosaniline) aqueous solution and 20 mL of 1 M HCl. Sodium metabisulphite (0.1 g) was added to every 6 mL of Schiff reagent before use, and the resultant solution was incubated at 37°C until it became colorless or pale yellow. Periodic acid reagent was freshly prepared by adding 10 μL of 50% periodic acid solution to 7 mL of 7% (v/v) acetic acid solution.

Standard calibration curves were prepared from 2 mL of mucin standard solutions (0.1, 0.2, 0.3, 0.4 and 0.5 mg/2 mL). After adding 0.2 mL of periodic acid reagent, the solutions were incubated at 37°C for 2 h in a water bath. Then, 0.2 mL of Schiff reagent was added at room temperature. After 30 min, the absorbance of the solution was recorded at 555 nm by a UV spectrophotometer (He et al., 1998) (Shimadzu UV1610, Japan). All the samples were determined with the same procedure. The mucin content was calculated from the standard calibration curve. Each experiment was performed in triplicate and average values were calculated.

Adsorption of mucin to chitosan microspheres

Mucin aqueous solution with different concentrations (0.1, 0.2, 0.4, 0.8 and 2.0 mg/mL) were prepared. Chitosan microspheres (5 mg) were dispersed in the above mucin solutions (2 mL), vortexed, and shaken at room temperature for 1 h (He et al., 1998). Then, the dispersions were centrifuged at 4000 rpm for 4 min, and the supernatant was used for the measurement of the mucin content by mucus glycoprotein assay. The mucin concentration was calculated from a calibration curve, and the amount of mucin adsorbed to the microspheres was calculated as the difference between the total amount of mucin added and the free mucin in the supernatant. The data obtained were interpreted using Freundlich and Langmuir equations describing the adsorption isotherms.

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

Measurement of *in vitro* mucoadhesion

The *in vitro* mucoadhesion of microspheres was carried out by modifying the method described by Ranga Rao et al. and others (Ranga Rao and Buri, 1989; Patil and Murthy, 2006; Patil and Sawant, 2009) using sheep nasal mucosa. The microspheres were placed on sheep nasal mucosa after fixing to the polyethylene support. The mucosa was then placed in the desiccator to maintain at > 80% RH at room temperature for 30 min to allow the polymer to hydrate and to prevent drying of the mucus. The mucosa was then observed under microscope, and the number of particles attached to the particular area was counted. After 30 min, the polyethylene support was introduced into a plastic tube cut in circular manner and held in an inclined position at an angle of 45°. Mucosa was washed thoroughly for 5 min with phosphate buffer pH 6.2. Tissue was again observed under microscope to see the number of microspheres remaining in the same field area.

The adhesion number was determined by the following equation:

$$\mathrm{Na} = \frac{\mathrm{N}}{\mathrm{N}_0} \times 100,$$

where Na is adhesion number, N_0 is total number of particles in a particular area, and N is number of particles attached to the mucosa after washing.

***In vitro* drug release**

The drug release profiles of CRV loaded chitosan microspheres were evaluated using a Franz diffusion cell apparatus which consisted of donor and receptor compartments. A dialysis membrane (pretreated) (cut-off Mw 12000, Hi Media, India) was used to keep the sprinkled microspheres on the donor side which allowed free diffusion of CRV to the receptor compartment containing 20 mL phosphate buffer solution pH 6.2 (within the pH range in nasal cavity) and maintained at 37 ± 1°C. The receptor compartment was stirred with a magnetic stirring bar. At scheduled time intervals, aliquots were withdrawn from receptor cell and replaced with same volume of fresh medium. The samples were assayed spectrophotometrically at 242 nm. All experiments were carried out in triplicate and average values were calculated.

DSC analysis

DSC (Mettler Toledo DSC 822) was performed on placebo chitosan microspheres, drug-loaded chitosan microspheres and plain CRV to determine the thermal behavior. It was also used to determine the existence of possible interaction between the polymer and drug. Samples were heated from 30°C to 300°C at the heating rate of 10°C/min in nitrogen atmosphere (flow rate, 20 mL/min).

XRD studies

The XRD patterns of placebo chitosan microspheres, drug-loaded microspheres and plain CRV were recorded using X-ray diffractometer (Bruker AXS D8 Advance, with X-ray source of Cu, Wavelength 1.5406 Å and Si (Li) PSD detector). These studies are useful to investigate crystallinity of drug in the microspheres. The samples were mounted on a sample holder and XRD patterns were recorded in the range of 3–50° at the speed of 5°/min.

Radiolabeling and animal studies

Radiolabeling of microspheres. Microspheres (CHC6) were labeled with technetium-99m (^{99m}Tc) by direct labeling method as described by Tafaghodi et al (Tafaghodi et al., 2004). The radiolabeling procedure was carried out in the presence of the powerful reducing agent, stannous chloride. The stannous ion reduces 99m-technetium from the +7 oxidation state to the more reactive +5 oxidation state to promote binding. Ten milligrams of microspheres were suspended in the labeling medium containing 0.5 mL of normal saline, 50 μL stannous chloride (5 mg/mL) and 1 mL technetium–99m pertechnetate eluate containing about 3 mCi of activity and pH was adjusted to 6.5 using 0.5 M sodium bicarbonate. The mixture was left under continuous stirring for about 10 min and separated by centrifugation. Microspheres were washed with 2 × 5 mL sterile distilled water and supernatants were collected. The labeled microspheres were washed with acetone (2 × 5 mL). Microspheres were separated by centrifugation and dried by incubation at 60°C for 30 min.

Radiolabeling of CRV. CRV was labeled with technetium-99m (^{99m}Tc) by direct labeling method. Briefly, to 1 mL of drug solution (2 mg/mL) mixture containing 40 μL stannous chloride (5 mg/mL) and 0.5 mL technetium–99m pertechnetate eluate containing about 1 mCi of activity was added and pH was adjusted to 7.4 using 0.5 M sodium bicarbonate. Then mixture was incubated for 30 min at room temperature.

Determination labeling efficiency of CRV and microspheres. The labeling efficiency was determined by ascending instant thin layer chromatography (ITLC) using silica gel (SG) coated fiber sheets (Gelman Sciences Inc, Ann Arbor, MI). The ITLC was performed using 100% acetone or 0.9% saline as the mobile phase. Around 2–3 μL of the radiolabeled complex (in case of microspheres before washing step) was applied at a point 1 cm from the end of an ITLC-SG strip. The strip was developed in acetone or 0.9% saline, and the solvent front was allowed to reach at the top. The strip was cut into two halves, and the radioactivity in each segment was measured in a shielded well-type gamma scintillation counter (Caprac-R, Capintec, USA).

RIGHTSLINK

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

The radiolabeling efficiency was evaluated with ITLC-SG strips as stationary phase and acetone 100% as the mobile phase.

$$\%\,\text{Radiolabelling} = \frac{\text{Radioactivity (counts) retained in the lower half of the strip}}{\text{Initial radioactivity associated (total count present) with the strip}} \times 100$$

In vivo studies

The *in vivo* studies were performed following the guidelines approved by the Committee for the Purpose of Control and Supervision of Experiments on Animals, Ministry of Social Justice and Empowerment, Government of India. The animal protocol was duly approved by the Institutional Animal Ethics Committee.

Twelve New Zealand white rabbits weighing 2.8 ± 0.4 kg were divided in two groups. The animals were fasted overnight prior to the experiment, with free access to water. To one group of rabbits, radiolabeled microspheres (CHC6) (approximately 5–7 mg) were administered intranasally using monodose insufflator (Miat, Milano, Italy). To other group, radiolabeled CRV was administered intravenously (0.5 mL). At selected time intervals, blood samples were withdrawn from the marginal ear vein of the rabbits. The radioactivity in terms of counts per minute (cpm) was measured in a well-type gamma scintillation counter (Caprac-R, Capintec, USA). The animals were conscious during the whole experiment and between each blood sampling they were allowed to move freely within an enclosed area.

Pharmacokinetics analysis

The noncompartmental pharmacokinetic analysis was performed by Kinetica 5.0 (Thermo Fisher Scientific, USA) and maximum plasma concentration (C_{max}), its time of occurrence (T_{max}) and the area under the curve (AUC) were determined from the individual time versus radioactivity profiles. The bioavailability (F) of the intranasal (IN) dose of microsphere formulation was calculated with the following equation:

$$F = \frac{AUC_{IN} \times Dose_{IV}}{AUC_{IV} \times Dose_{IN}} \times 100\%$$

Here AUC_{IN} and AUC_{IV} are the individual areas under radioactivity time curves of each rabbit administered microspheres containing CRV ($Dose_{IN}$) intranasally and that of the free CRV solution administered intravenously, respectively.

Gamma scintigraphy

The deposition, distribution and subsequent clearance of microspheres were studied by gamma scintigraphy and imaging was performed immediately at 0 min and 4 h post administration of microspheres to nasal cavity of rabbits using a Single Photon Emission Computerized Tomography, (LC 75–005, Diacam, Siemens AG, Earlangan, Germany) gamma camera. The quantification of the data was made defining region of interest, around the desirable area of the nasal cavity. The highest count rate at 0 min after dosing was assigned a 100% value, which was then used to calculate the percentage remaining for the other time point.

Statistical data analysis

The results of *in vitro* and *in vivo* experiments are reported as mean ± S.D. of mean ($n = 3$). Statistical tests of significance were performed using analysis of variance and Student *t* test and differences were considered to be statistically significant when $p < 0.05$.

Results and discussion

Particle size measurements

The particle size of the microspheres was in the range of 20.82–49.26 µm (Table 2), which is favorable for intranasal administration. At lower concentration of chitosan (1%), the microspheres with size 15–20 µm were obtained but aggregation and clumping was observed. At higher concentration (3%), the chitosan solution was so viscous that it was difficult to pass through syringe and yielded mean particle size of 60.48 µm. This significant increase may be because of the increased viscosity of the chitosan solution, resulting in difficulty in dispersion and subdivision of droplets (Aiedeh et al., 1997). At medium concentration (2%), the mean particle size was observed to be 32.23 µm. Hence the optimized concentration of 2% was selected for preparing the different batches of the microspheres.

The size of the microspheres was found to increase with an increase in drug loading. This increase in size of the microspheres may be due to the increase in viscosity of the internal phase caused by the increase in drug concentration as explained by Denkbas et al. (Denkbas et al., 1999).

Morphological characterization

The microspheres were non aggregated, free flowing powders with spherical shape and smooth surface (Figure 1). However, one noticeable characteristic of the cross-linked chitosan microspheres was their yellow to brownish color. The intensity of the color of the microspheres increased as the volume of GA and time of cross-linking increased. Roberts and Taylor (Roberts and Taylor, 1989) reported that the chitosan/GA gels of yellow-brown color were produced due to inter-chain cross-link formation.

RIGHTSLINK

Table 2. Characteristics of prepared carvedilol loaded chitosan microspheres*.

Formulation code	Particle size (μm)	Angle of repose (θ)	Compressibility index (%)	Drug loading (%)	Entrapment efficiency (%)	*In vitro* mucoadhesion (%)
CHC1	24.44 ± 1.67	30.73 ± 1.90	16.29 ± 1.66	15.62 ± 0.86	47.45 ± 1.03	83.21 ± 2.21
CHC2	42.53 ± 1.38	28.81 ± 1.34	15.33 ± 1.15	21.47 ± 0.59	62.40 ± 1.50	82.89 ± 1.28
CHC3	20.52 ± 1.19	33.13 ± 1.08	18.43 ± 1.47	13.87 ± 0.94	43.31 ± 1.09	75.21 ± 1.44
CHC4	38.55 ± 1.10	31.40 ± 1.53	15.51 ± 0.72	19.35 ± 1.02	56.57 ± 0.55	74.74 ± 1.27
CHC5	34.86 ± 1.58	29.66 ± 0.93	15.61 ± 0.63	17.57 ± 0.73	52.35 ± 1.59	88.56 ± 1.74
CHC6	49.26 ± 1.89	32.08 ± 1.58	17.47 ± 1.07	23.24 ± 0.92	66.86 ± 1.77	80.27 ± 2.49
CHC7	33.11 ± 2.37	31.33 ± 1.02	15.85 ± 1.48	14.27 ± 1.22	41.25 ± 1.01	79.60 ± 3.15
CHC8	46.02 ± 1.67	33.32 ± 1.46	18.89 ± 1.21	18.65 ± 0.79	54.30 ± 1.23	74.72 ± 1.54

*Mean ± SD (n = 3)

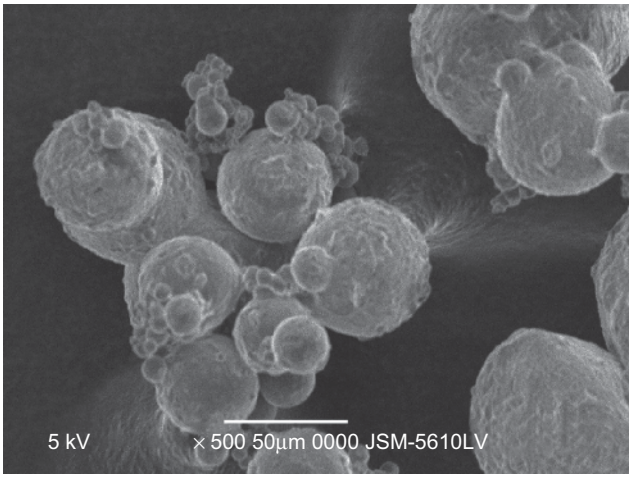

Figure 1. Scanning electron microscopy photograph of carvedilol (CRV) loaded chitosan microspheres.

Flow properties

The results of flow properties measurements are shown in Table 2. The values of angle of repose were in the range of 28.81° ± 1.34–33.32° ± 1.46 which is within the normal acceptable range of 20°–40° (Wells and Aulton, 1988). The microspheres thus showed reasonably good flow potential. The values of compressibility index which were in the range 15.33 ± 1.15–18.89 ± 1.21, also indicating good flow characteristics of the microspheres.

Entrapment efficiency

The percentage EE was found to be in the range between 42.62 and 67.24. The % EE was found to be proportional to drug loading. The formulations loaded with higher amount of drug exhibited higher entrapment efficiencies. The increase in drug loading may cause subsequent increase in viscosity of the polymer dispersion, which would decrease the probability of drug diffusion into the external phase during preparation, resulting in higher drug EE. However, with increasing amount of cross-linking agent, EE decreased (Table 2). This may be due to an increase in cross-link density that will thereby reduce the free spaces within the polymer matrix and hence lead to reduced EE.

Assessment of the mucoadhesive behavior of chitosan microspheres by mucus glycoprotein Assay

Mucoadhesion involves various types of interaction forces between mucoadhesive materials and mucus surface, such as hydrogen bonding, electrostatic attraction, van der Waals forces and mechanical interpenetration and entanglement. Many methods have been employed to evaluate these interactions *in vitro* and *in vivo*. Commercial mucin is frequently used as a substitute for fresh mucin because of its reproducible quality and easy availability. As a strong interaction exists between mucin and chitosan, mucin should be spontaneously adsorbed on the surface of the chitosan microspheres. Hence, the mucoadhesive property of chitosan microspheres was assessed by the suspension of chitosan microspheres in different amounts of mucin (Type III) in aqueous solutions at room temperature. The linearity range for mucin at the wavelength of detection of 555 nm was obtained as 0.05–0.25 mg/mL. The linear regression equation obtained by least square method was, $y = 1.978x + 0.4493$. As the mucin concentration was increased, the amount of mucin adsorbed increased. Thus, these studies indicate that mucoadhesion may increase the residence time of the formulation in the nasal cavity.

Adsorption isotherms

The data obtained from adsorption studies were fitted to Freundlich and Langmuir equations. Straight lines were obtained, and the constants (n, K in case of Freundlich isotherm and a, b for Langmuir isotherm) from these lines were calculated and are listed in Table 3. It was found that the values of R^2 were significantly higher ($p < 0.05$) for the Langmuir equation as compared to the Freundlich equation. This explains a more specific adsorption process where electrostatic interaction is involved. The adsorption of mucin to chitosan is expected to be dominated by the electrostatic attraction

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

between the positively charged chitosan (containing amino groups) and negatively charged mucin (containing ionized sialic acid).

In vitro mucoadhesion

The results of *in vitro* mucoadhesion (Table 2) showed that all the batches of microspheres had satisfactory mucoadhesive property ranging from 74.72 ± 1.54% to 88.56 ± 1.74% and could adequately adhere on nasal mucosa. The results also showed that with increasing polymer ratio, higher mucoadhesion percentages were obtained ($p < 0.05$). This may be due to the fact that, as the amount of polymer increased, the amino groups available for binding with the sialic acid residues in mucus layer also increase, resulting in increased mucoadhesion. The percent *in vitro* mucoadhesion was found to decrease slightly with increase in the volume of GA ($p < 0.05$). (Table 2), attributed to increased rigidity and reduced binding sites for mucoadhesion. Most of the studies showed that the prerequisite for a good mucoadhesion is the high flexibility of polymer backbone structure and of its polar functional groups. Such a flexibility of the polymer chains, however, is reduced if the polymer molecules are highly cross linked. Although highly cross-linked microspheres will absorb water, they are insoluble and will not form a liquid gel on the nasal epithelium but rather a more solid gel-like structure. This decrease in flexibility imposed upon polymer chains by the cross-linking agent makes it more difficult for highly cross-linked polymers to penetrate the mucin network (Berger et al., 2004). Thus, highly cross-linked microparticles effectively limits the length of polymer chains that can penetrate the mucus layer, could possibly decrease the mucoadhesion strength of the microspheres (Illum et al., 1987).

In vitro drug release

The drug release behavior of the microspheres is shown in Figure 2. The release pattern showed a moderate and controlled release following near zero order release. In order to understand the mechanism and kinetics of drug release, the data was analyzed by Peppas equation (Ritger and Peppas, 1987), $Mt/M\infty = kt^n$, where Mt is the amount of drug released at time t, M∞ is the amount released at time ∞, Mt/M∞ is the fraction of drug released at time t, k is a constant characteristic of the drug-polymer system and n is the diffusional exponent, a measure of the primary mechanism of drug release. Using the least squares procedure, the values of n, k and correlation coefficient (r) were estimated.

Table 3. Constants for Langmuir and Freundlich Equations.

	Freundlich isotherm[a]			Langmuir isotherm[b]		
Batch code	n	K	R^2	a	b	R^2
CHC1	0.934	2.65	0.9785	1.0628	3.5150	0.9885
CHC2	0.922	2.77	0.9692	0.6178	3.0985	0.9909
CHC3	1.07	3.29	0.9727	0.9946	3.5609	0.9903
CHC4	0.88	2.71	0.9585	1.6925	4.1211	0.9807
CHC5	1.03	3.75	0.9758	0.7629	4.1364	0.9894
CHC6	0.86	2.49	0.9781	0.2152	3.4444	0.9845
CHC7	1.01	2.34	0.9427	0.0462	3.2795	0.9895
CHC8	0.98	2.95	0.9715	0.9089	3.7022	0.9841

[a] $x/m = K \cdot Ce^{1/n}$; [b] $1/(x/m) = a + b \cdot 1/Ce$.
x/m is mucin adsorbed, K and n are Freundlich isotherm constants, a and b are Langmuir isotherm constants, and Ce is the free mucin.

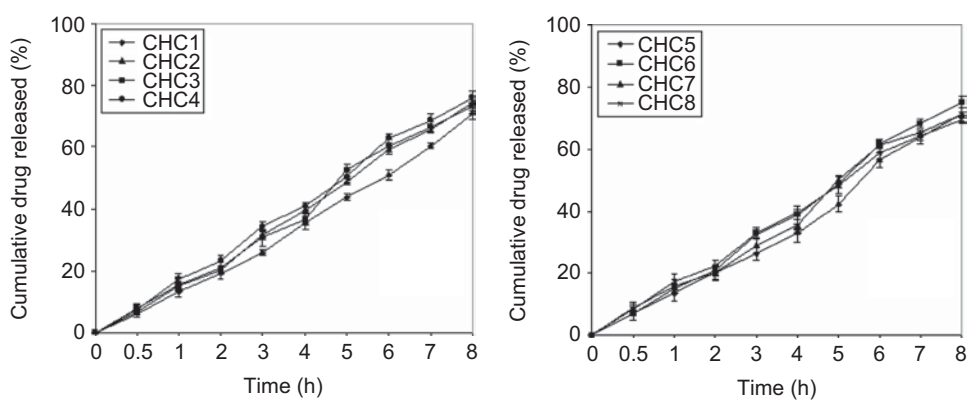

Figure 2. In *vitro* drug release profile of chitosan microspheres of carvedilol (CRV). The values are mean ± S.D. (n = 3).

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

In spherical matrices, if $n \leq 0.43$, a Fickian diffusion (case-I), $0.43 \leq n < 0.85$, anomalous or non-Fickian transport and $n \geq 0.85$, a case-II transport (zero order) drug release mechanism dominates. The values of n for all the batches ranged from 0.74 to 0.84 with correlation coefficient close to 0.99, indicating non-Fickian or anomalous type of transport. Non- Fickian release is described by two mechanisms: a combination of drug diffusion and polymer relaxation. The release mechanism is known to be influenced by (a) non homogeneous gel microstructure as well as the existence of polymeric domains within the swollen gel, (b) rate of fluid ingress into the matrix, (c) dissociation/ erosion and total disentanglement at the dissolution front and (d) rate of matrix swelling, relaxation as well as molecular diffusion of drug through the swollen gel.

Formulation CHC6 showed EE of 66.86 ± 1.77% with particle size of 49.26 ± 1.89 µm and *in vitro* mucoadhesion of 80.27 ± 2.49% with around 75% release of drug in 8 h. Hence it was selected as optimized formulation for further studies including *in vivo* evaluation.

Differential scanning calorimetry

DSC thermograms of pure CRV (a), CRV loaded chitosan microspheres (b) and blank chitosan microspheres (c) are displayed in Figure 3. Thermogram of CRV showed a sharp peak at 122°C, indicating the melting of the drug. In case of blank microspheres, a broad peak was observed at 96°C was mainly due to water evaporation. The transitions associated with loss of water correspond to the hydrophilic nature of the functional groups of the polymer. The second thermal event registered was an exothermic peak occurring at 273°C which can express the overall exothermic effect connected with decomposition. Similarly in case of drug loaded microspheres, a broad endothermic peak at 76°C and exothermic peak at 274°C was observed. However, there was no peak corresponding to CRV, indicating that CRV is molecularly dispersed in the matrix.

XRD studies

XRD patterns recorded for plain CRV (a), placebo microspheres (b) and CRV loaded microspheres (c) are presented in Figure 4. CRV peaks observed at 2θ of 12°, 14°, 18° and 25° are due to the crystalline nature of CRV. These peaks were not observed in the CRV loaded microspheres. This indicates that drug particles are dispersed at molecular level in the polymer matrices since no indication about the crystalline nature of the drugs was observed in the drug loaded microspheres.

Labeling efficiency of microspheres and CRV

Microspheres and CRV were labeled by the direct labeling method using reduced ^{99m}Tc. Results for radiolabeling efficiency and stability of the ^{99m}Tc labeled complex were obtained by ITLC using SG coated fiber sheets using

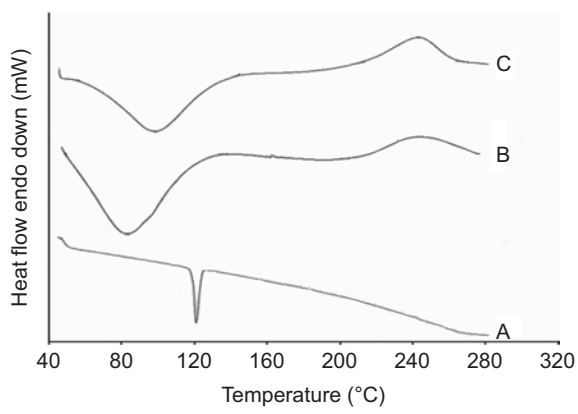

Figure 3. Differential scanning calorimetry (DSC) thermograms of (A) pure carvedilol (CRV); (B) drug loaded microspheres; and (C) placebo microspheres.

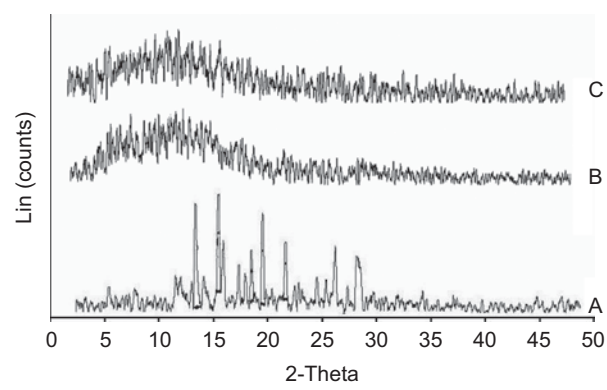

Figure 4. Powder X-ray diffraction (XRD) patterns of (A) pure carvedilol; (B) drug loaded microspheres; and (C) placebo microspheres.

Table 4. Pharmacokinetic parameters following intranasal and intravenous administration.

Formulation	T_{max} (h)	C_{max} (kcpm/g)	AUC_{0-8h} (kcpm/h/g)*	$T_{1/2}$ (h)	MRT (h)	K_{el} (h^{-1})	Bioavailability (F, %)
IV (CRV) (0.5 mg/kg)	–	–	54.06 ± 6.45	1.99 ± 0.32	2.18 ± 0.78	0.346 ± 0.092	100
Microspheres (3 mg/kg)	1	78.79	236.59 ± 21.68#	2.93 ± 1.24	4.31 ± 1.02	0.236 ± 0.076	72.29

*Mean ± S.D. (n = 4); #$p < 0.05$.

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

saline or 100% acetone as the solvent. Acceptably high labeling efficiency was found for microspheres and CRV (96.36 ± 2.05 and 98.45 ± 0.97% respectively).

In vivo studies

The relevant pharmacokinetic parameters including maximum concentration (C_{max}), time of maximum plasma concentration (T_{max}), the AUC, plasma half life ($T_{1/2}$), mean residence time, elimination rate constant (K_{el}) and relative bioavailability are shown in Table 4. Blood kinetics data of CRV and the microsphere formulation at various time intervals are shown in Figure 5. The AUC after intranasal administration of CRV and microspheres were about 236.59 ± 21.68 KCPM/h/gm and 54.06 ± 6.45 KCPM/h/gm respectively which was statistically significant ($p < 0.05$; Student's *t* test).

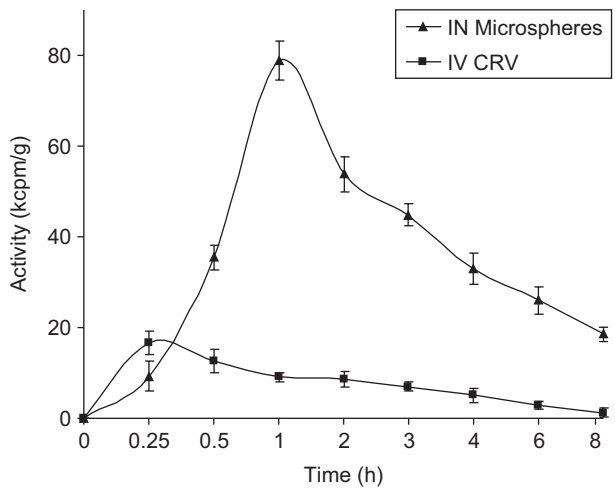

Figure 5. Blood radioactivity time profiles of carvedilol (CRV) after administration of microspheres intranasally (IN) (3 mg/kg) and CRV solution intravenously (IV) (0.5 mg/kg) in rabbits.

The nasal bioavailability (*F*) was found to be 72.29% indicating that nasal administration results in improved absorption of CRV from chitosan microspheres in rabbits. Illum et al. (Illum et al., 2001) described bioadhesive starch microspheres for nasal administration which were retained in the nasal cavity for an extended time period owing to their bioadhesive nature and from which, after gelling, a local high drug concentration was reached. It is reported in previous studies (Illum, 2003; Alpar et al., 2005; Gavini et al., 2005) that chitosan has absorption enhancing effect, as it improves the paracellular transport by opening the tight junctions. The high CRV absorption through nasal mucosa may be attributed to the combined effects of bioadhesion and absorption enhancement due to chitosan. It has been demonstrated that the mucoadhesive microparticles have a significant effect on the mucosal uptake of drugs (Pereswetoff-Morath, 1998; Ugwoke et al., 2001). Hence, both the higher local drug concentration and the increased paracellular transport are likely to play an important role in absorption process.

Gamma scintigraphy

The gamma scintigraphy images showed that the microsphere powder was spread over a wide area within the nasal cavity of rabbits (Figure 6). The scintigraphy images showed chitosan dependent progressive migration of radioactivity with time from the nasal cavity to the stomach and intestine. The activity determined for RIO at 0 min after administration was considered 100%. The activity after 4 h post administration was found 38.45%. These results indicated that the microspheres cleared slowly and were retained for extended periods in the nasal cavity, thereby providing sustained and enhanced drug absorption from the nasal mucosa, as confirmed from pharmacokinetic studies.

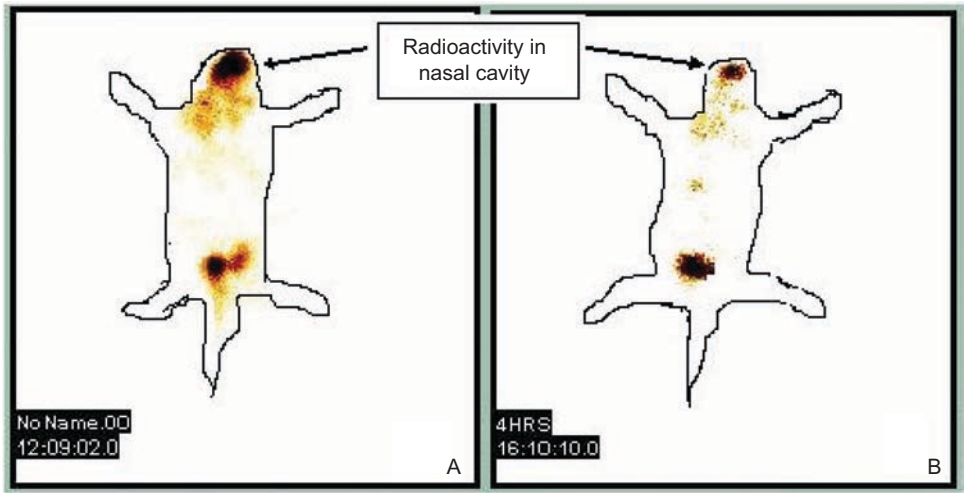

Figure 6. Scintigraphy image of rabbit at 0 min (A) and 4 h (B) post administration of microspheres to nasal cavity.

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

Conclusion

Free flowing chitosan microspheres of CRV with a smooth surface and spherical shape were prepared by emulsification cross-linking method. The size of the microspheres was in the range of 20–50 μm, which is favorable for intranasal absorption. The EE was observed in the range of 42–68%. All the batches had good mucoadhesive property ranging from 74% to 88%. A strong interaction between mucin and chitosan microspheres was detected which explains a more specific adsorption process where electrostatic interaction between the positively charged chitosan and negatively charged mucin is involved. All batches of microspheres released around 75% of drug in 8 h. DSC and XRD studies revealed that CRV was molecularly dispersed in the polymer matrix. The pharmacokinetics of chitosan microspheres after nasal administration in rabbits showed that the microspheres were able to promote enhanced drug absorption through the nasal mucosa and to remarkably improve the bioavailability of CRV.

Acknowledgements

Financial assistance to Mr. Sanjay B. Patil in terms of Quality Improvement Programme by All India Council for Technical Education, New Delhi, India is duly acknowledged. The authors are thankful to Dr. R. P. Thripathi, Director, Institute of Nuclear Medicine and Allied Sciences (INMAS), Delhi, for providing necessary facilities to carry out the radiolabeling experiments. Technical assistance during radiolabeling studies by Mrs. Ankur Kaul, INMAS, Delhi is highly acknowledged. The authors are thankful to Miat, Milano, Italy for providing gift sample of monodose powder nasal insufflator.

***Declaration of interest*:** The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

References

Aiedeh K, Gianasi E, Orienti I, Zecchi V. (1997). Chitosan microcapsules as controlled release systems for insulin. J Microencapsul, 14, 567–576.

Alpar HO, Somavarapu S, Atuah KN, Bramwell VW. (2005). Biodegradable mucoadhesive particulates for nasal and pulmonary antigen and DNA delivery. Adv Drug Deliv Rev, 57, 411–430.

Banker GS, Anderson NR. (1991). In: Lachman L, Lieberman HA, Kanig J, eds. The Theory and Practice Of Industrial Pharmacy pp 317. Mumbai, Maharashtra, India: Varghese publishing house.

Berger J, Reist M, Mayer JM, Felt O, Peppas NA, Gurny R. (2004). Structure and interactions in covalently and ionically crosslinked chitosan hydrogels for biomedical applications. Eur J Pharm Biopharm, 57, 19–34.

Bernkop-Schnürch A, Hornof M, Zoidl T. (2003). Thiolated polymers–thiomers: synthesis and *in vitro* evaluation of chitosan-2-iminothiolane conjugates. Int J Pharm, 260, 229–237.

Berthod F, Saintigny G, Chretien F, Hayek D, Collombel C, Damour O. (1994). Optimization of thickness, pore size and mechanical properties of a biomaterial designed for deep burn coverage. Clin Mater, 15, 259–265.

Blanco MD, Gómez C, Olmo R, Muñiz E, Teijón JM. (2000). Chitosan microspheres in PLG films as devices for cytarabine release. Int J Pharm, 202, 29–39.

Chien YW, Su KSE, Chang SF. (1989). Anatomy and physiology of the nose. In: Chien YW, Su KSE, Chang SF, eds. Nasal Systemic Drug Delivery 39, pp 1–19. New York: Marcel Dekker.

Cornaz A, Buri P. (1994). Nasal mucosa as an absorption barrier. Eur J Pharm Biopharm, 40, 261–270.

Denkbas EB, Seyyal M, Piskin E. (1999). 5-fluorouracil loaded chitosan microspheres for chemoembolization. J Microencapsul, 16, 741–749.

Fernandez-Urrusuno R, Romani D, Calvo P, Vila-Jato JL, Alonso MJ. (1999). Development of a freeze-dried formulation of insulin-loaded chitosan nanoparticles intended for nasal administration. STP Pharma Sciences, 9, 429–436.

Gavini E, Rassu G, Muzzarelli C, Cossu M, Giunchedi P. (2008). Spray-dried microspheres based on methylpyrrolidinone chitosan as new carrier for nasal administration of metoclopramide. Eur J Pharm Biopharm, 68, 245–252.

Gavini E, Rassu G, Sanna V, Cossu M, Giunchedi P. (2005). Mucoadhesive microspheres for nasal administration of an antiemetic drug, metoclopramide: in-vitro/ex-vivo studies. J Pharm Pharmacol, 57, 287–294.

He P, Davis SS, Illum L. (1998). In vitro evaluation of the mucoadhesive properties of chitosan microspheres. Int J Pharm, 166, 75–88.

Illum L. (1998). Bioadhesive formulations for nasal peptide delivery. In: Mathiowitz E, Lehr CM, Chickering D, eds. Bioadhesion in Drug Delivery - Issues in Fundamentals, Novel Approaches and Development pp 507–539. New York: Marcel Dekker.

Illum L. (2002). Nasal drug delivery: new developments and strategies. Drug Discov Today, 7, 1184–1189.

Illum L. (2003). Nasal drug delivery-possibilities, problems and solutions. J Control Rel, 87, 187–198.

Illum L, Jorgensen H, Bisgaard H, Krogsgaard O, Rossing N. (1987). Bioadhesive microspheres as a potential nasal drug delivery system. Int J Pharm, 39, 189–199.

Illum L, Fisher AN, Jabbal-Gill I, Davis SS. (2001). Bioadhesive starch microspheres and absorption enhancing agents act synergistically to enhance the nasal absorption of polypeptides. Int J Pharm, 222, 109–119.

Mantle M, Allen A. (1978). A colorimetric assay for glycoproteins based on the periodic acid/Schiff stain [proceedings]. Biochem Soc Trans, 6, 607–609.

Mi FL, Wong TB, Shyu SS, Chang SF. 1999. Chitosan microspheres: modification of polymeric chem-physical properties of spray-dried microspheres to control the release of antibiotic drug. J Applied Polym Science, 71, 747–759.

Mygind N, Dahl R. 1998. Anatomy, physiology and function of the nasal cavity. Adv Drug Deliv Rev, 29, 3–12.

Packer M, Fowler MB, Roecker EB, Coats AJ, Katus HA, Krum H, Mohacsi P, Rouleau JL, Tendera M, Staiger C, Holcslaw TL, Amann-Zalan I, DeMets DL. (2002). Effect of carvedilol on the morbidity of patients with severe chronic heart failure: results of the carvedilol prospective randomized cumulative survival (COPERNICUS) study. Circulation, 106, 2194–2199.

Patil SB, Murthy RSR. (2006). Preparation and in vitro evaluation of mucoadhesive chitosan microspheres of amlodipine besylate for nasal administration. Ind J Pharm Sci, 8, 64–67.

Patil SB, Sawant KK. (2009). Development, optimization and in vitro evaluation of alginate mucoadhesive microspheres of carvedilol for nasal delivery. J Microencapsul, 26, 432–443.

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK

Pereswetoff-Morath L. (1998). Microspheres as nasal drug delivery systems. Adv Drug Deliv Rev, 29, 185–194.
Ranga Rao KV, Buri P. (1989). A novel situ method to test polymer and coated microparticles for bioadhesion. Int J Pharm, 52, 265–270.
Ritger PL, Peppas NA. (1987). A simple equation for description of solute release II. Fickian and anomalous release from swellable devices. J Control Rel, 5, 37–42.
Roberts GAF, Taylor KE. (1989). The formation of gels by reaction of chitosan with glutaraldehyde. Macromolecular Chemistry, 190, 951–960.
Soane RJ, Frier M, Perkins AC, Jones NS, Davis SS, Illum L. (1999). Evaluation of the clearance characteristics of bioadhesive systems in humans. Int J Pharm, 178, 55–65.
Tafaghodi M, Abolghasem Sajadi Tabassi S, Jaafari MR, Zakavi SR, Momen-Nejad M. (2004). Evaluation of the clearance characteristics of various microspheres in the human nose by gamma-scintigraphy. Int J Pharm, 280, 125–135.
Taylor MK, Ginsburg J, Hickey AJ, Gheyas F. (2000). Composite method to quantify powder flow as a screening method in early t or capsule formulation development. AAPS Pharm Sci Tech, 1, Article 18.
Thanoo BC, Sunny MC, Jayakrishnan A. (1992). Cross-linked chitosan microspheres: preparation and evaluation as a matrix for the controlled release of pharmaceuticals. J Pharm Pharmacol, 44, 283–286.
Thummel KE, Shen DD. (2001). Design and optimization of dosage regimens: pharmacokinetic data. In: Hardman JG, Limbirel LE, Gilman AG, eds. Goodman and Gilman's The Pharmacological Basis of Therapeutics pp 1936–1940. New York: McGraw Hill.
Ikechukwu Ugwoke M, Kaufmann G, Verbeke N, Kinget R. (2000). Intranasal bioavailability of apomorphine from carboxymethylcellulose-based drug delivery systems. Int J Pharm, 202, 125–131.
Ugwoke MI, Verbeke N, Kinget R. (2001). The biopharmaceutical aspects of nasal mucoadhesive drug delivery. J Pharm Pharmacol, 53, 3–21.
van der Lubben IM, Kersten G, Fretz MM, Beuvery C, Coos Verhoef J, Junginger HE. (2003). Chitosan microparticles for mucosal vaccination against diphtheria: oral and nasal efficacy studies in mice. Vaccine, 21, 1400–1408.
Vidgren P, Vidgren M, Vainio P, Nuutinen J, Paronen P. (1991). Double-labeling technique in the evaluation of nasal mucoadhesion of disodium cromoglycate microspheres. Int J Pharm, 73, 131–136.
Wang YM, Sato H, Adachi I, Horikoshi I. (1996). Optimization of the formulation design of chitosan microspheres containing cisplatin. J Pharm Sci, 85, 1204–1210.
Wells J, Aulton M. (1988). Preformulation. In: Aulton ME, ed. Pharmaceutics: The Science of Dosage Form Design pp 223–253. Edinburgh: Churchill Livingstone.

Journal of Drug Targeting Downloaded from informahealthcare.com by Dalhousie University on 12/04/12
For personal use only.

RIGHTSLINK